SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	Sasikumar Megha	**Reg. No.:**	192471038
Section / Batch:	3rd, CS&BS	**Date:**	20/07/2026

Code of Conduct

I Sasikumar Megha *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: S. Megha

Instructions

- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
		GRAND TOTAL:	100 Marks

38	192524251	VAISHNAV M	1. Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow. 2. A stack has a capacity of 5 elements. Explain what happens if the following operations are executed: Push(A), Push(B), Push(C), Push(D), Push(E), Push(F). Identify the error and suggest two possible solutions.
39	192525187	VEERA SANTHOSH A	1. Compare the performance of a linear queue and a circular queue when 100 enqueue and dequeue operations are performed alternately. 2. A queue implemented using an array report "Queue Full" even though some positions are empty. ➤ Analyze why this occurs. ➤ Explain how a circular queue solves this problem.
40	192511226	VINIL KUMAR S	1. Consider the singly linked list: 10 → 20 → 30 → 40 → 50 Delete the node containing 30. Explain how the links change after deletion. 2. Given two sorted linked lists: L1: 1 → 3 → 5 → 7 L2: 2 → 4 → 6 → 8

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

Linked stack operations

A stack follows the LIFO (Last In, First Out) principle. In a linked stack, insertion and deletion are performed at the top.

The elements 100, 200, 300, 400 & 500 are inserted into the stack.

500
↓
400
↓
300
↓
200
↓
100

* Here 500 is the Top because it was inserted Last.

→ If we perform the normal POP() operation, the elements are removed from the TOP.

∴ The first elements removed will be

500 → 400

Hence 100 and 200 cannot be directly popped from a normal stack while 500, 400 and 300 are still Present above them

if 100 and 200 are removed after accessing the lower nodes, the remaining stack structure will be:

Linked stack after Removing 100 and 200.

500
↓
400
↓
300

Conclusion: A linked stack allows deletion only from the top. Therefore 100, 200 cannot be directly popped first from the given stack.

2) Online food delivery Service - Queue

The orders received as

O1 → O2 → O3 → O4

while O2 is being prepared, an express order O5 arrives.

Why is a Normal Queue Not suitable?

A normal queue follows the FIFO (First in, first out) Principle. Therefore, the orders would normally be Processed as

01 → 02 → 03 → 04 → 05

Even though 05 is an express order, it would have to wait unit 03 and 04 are completed. This can cause unnecessary delays for the express customer.

Recommeded Queue Implementation:

A priority Queue is the most suitable ~~environment~~. implementation.

→ In priority queue, each order is assigned a Priority. Express order are given higher priority than normal orders.

Ex:

Normal order : 01, 02, 03, 04

Express order : 05 (high priority)

After O_2 is completed, O_5 can be processed before O_3 and O_4

$O_1 \rightarrow O_2 \rightarrow O_5 \rightarrow O_3 \rightarrow O_4$

Justification:

A priority queue is appropriate because it allows high priority orders to be processed before normal orders. This ensures that express food orders are delivered faster and customer requirements are satisfied.